

point scale. Of course, trying to draw these sorts of conclusions is fraught with difficulties. It could be that social activity does not cause happiness, but rather that happiness causes social activity; maybe happy people are more often in the jovial mood that leads to calling their friends and arranging social meetings. Disentangling correlation from causation is beyond the scope of this chapter, but regardless of the direction of causation, our simple decision tree has at least given us the fact of the association, which we can investigate further if we care to. As cartoonist Randall Munroe put it, “Correlation doesn’t imply causation, but it does waggle its eyebrows suggestively and gesture furtively while mouthing ‘look over there.’”

We know how to make a simple decision tree with two branches. Now we just need to perfect how we create branches and then make many of them for a better, more complete decision tree.

Smarter Splitting

When we compared the happiness levels of people with active versus inactive social lives, we used 5 as our *split point*, saying that those who were rated higher than 5 had an active social life and those who were rated at 5 or below had an inactive social life. We chose 5 because it is a natural middle point for ratings that go from 1 to 10. However, remember that our goal is to build an accurate predictor of happiness. Rather than splitting based on intuitions about what a natural midpoint is or what seems like an active social life, it would be best to make our binary split in some place that leads to the best possible accuracy.

In machine learning problems, there are a few different ways to measure accuracy. The most natural way is to find the sum of our errors. In our case, the error that interests us is the difference between our prediction of someone’s happiness rating and their actual happiness rating. If our decision tree predicts that your happiness is 6 but it’s actually 8, then that tree’s error for your rating is 2. If we add up the prediction errors for every respondent in some group, we can get an error sum that measures the decision tree’s accuracy for predicting the happiness of members of that group. The closer we can get our error sum to zero, the better our tree is (but please see “The Problem of Overfitting” on page 179 for important caveats). This snippet shows a simple way to find the error sum: